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Optimizing Stimulation Efficiency in Large Volume Foamed Acidizing Treatments in Tight Carbonate Formations

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Abstract

Block 5, located in the northern region of the Sultanate of Oman, is a mature oilfield that continues to face challenges in sustaining economic production levels despite the application of horizontal well technology to maximize reservoir contact. While drilling campaigns have delivered mechanical success, a significant proportion of wells remain underperforming due to the inherently tight nature of the carbonate formations and damage associated with completion and workover operations. Stimulation is therefore essential to restore productivity; however, the complexity of treating long open-hole horizontal sections in tight formations has historically necessitated high capital expenditure solutions.

To address these challenges, Large-Volume Acidizing (LVA) with foamed diversion has been adopted as a cost-effective alternative to conventional multistage stimulation systems, eliminating the need for coiled tubing intervention and complex completion hardware. Since its implementation in 2017, the LVA program has undergone progressive optimization in acid type, diversion techniques, and placement strategies, supported by advanced stimulation modeling.

Recent advancements in the 2025 campaign include the introduction of a next-generation Single-Phase Retarded Acid (NG-SPRA) system, providing enhanced wormhole propagation, extended thermal stability, and reduced frictional pressure losses. In parallel, high-frequency pressure monitoring (HFPM) was incorporated into the treatment workflow to enable evaluation of diversion effectiveness, cluster coverage, and acid placement, allowing dynamic operational adjustments without additional intervention complexity.

Field implementation of this integrated approach has demonstrated significant benefits. Initial post-stimulation oil rates exceeded three to four times the pre-treatment levels, with normalized productivity indices surpassing those achieved with previous-generation acid systems under comparable reservoir conditions. The combined NG-SPRA and HFPM methodology has delivered up to 85% cost savings relative to multistage completion alternatives while improving stimulation efficiency and lateral coverage. The

results from Block 5 illustrate a transferable, field-proven workflow for cost-efficient carbonate stimulation in long-reach horizontal wells, offering applicable best practices for similar reservoirs within Oman and analogous carbonate settings globally.

Introduction

Block 5, located in northwestern Oman, is a mature hydrocarbon province where recent development strategies have increasingly focused on unlocking remaining potential from tight carbonate reservoirs. Horizontal drilling has been widely implemented to extend reservoir contact and delay water breakthrough; however, well performance remains variable. Even wells optimally positioned from a geological and structural perspective often underperform due to low matrix permeability, the absence of pervasive natural fracturing, and near-wellbore damage incurred during drilling and completion operations (Mufargi et al., 2025).

The field's principal producing intervals—Shammar, Natih, and Shuaiba—each present distinct reservoir characteristics that directly influence stimulation design. The Shammar unit is a thin, dense carbonate with limited net pay. The Natih Formation, the most extensively developed horizon, exhibits complex interbedding of porous and tight layers, requiring accurate acid placement and diversion control. The Shuaiba Formation, although laterally homogeneous, possesses an extremely tight matrix that demands deep acid penetration for sustained productivity. The combination of vertical heterogeneity and lateral variation across these formations makes achieving uniform stimulation coverage in long horizontal sections a significant technical challenge.

Operational constraints add further complexity. Coiled tubing (CT) intervention is generally avoided to minimize cost, eliminating the option for mechanically segmented acid placement. Most wells are completed as long open-hole laterals—typically 1,000 to 1,500 m in length—making uniform fluid distribution difficult. In response, a Large-Volume Acidizing (LVA) methodology has been adopted, employing high-rate bullheading with engineered foamed diversion to improve coverage without CT conveyance. Over successive campaigns, the approach has been enhanced through the integration of advanced acid systems—most recently a next-generation Single-Phase Retarded Acid (NG-SPRA) per SPE-217841—and degradable diverters to improve placement efficiency and acid penetration.

A significant recent enhancement has been the incorporation of High-Frequency Pressure Monitoring (HFPM) into the stimulation workflow. This technology enables evaluation of diversion effectiveness and acid placement distribution without additional downhole intervention. In this initial pilot application, HFPM was deployed primarily as a diagnostic tool to evaluate the effectiveness of the stimulation campaign and validate diversion performance. The high-resolution pressure data provided a detailed understanding of fluid distribution and diversion efficiency. While dynamic adjustment of pumping schedules and diversion strategy was not implemented during this pilot, the insights gained have established the basis for its future operational use. In upcoming campaigns, HFPM is planned to be fully integrated into the execution workflow, enabling real-time optimization of pumping parameters and diversion sequences during treatment to enhance stage coverage and overall stimulation efficiency.

This paper builds on the work presented in SPE-224978-MS, which outlined the evolution of LVA treatments in Block 5 up to 2024. It serves as the second part of that study, updating the field's stimulation program with results from the 2025 campaign and highlighting recent technological advancements—including the deployment of NG-SPRA acid systems and the first pilot application of High-Frequency Pressure Monitoring (HFPM) for open-hole LVA treatments. These additions represent a significant step toward a more data-driven, adaptive stimulation workflow, enabling enhanced lateral coverage and improved reservoir contact in long, tight carbonate wells.

Field Background and Reservoir Challenges

Geological Context and Reservoir Characteristics

Block 5, located in northern Oman, occupies the northeastern margin of the Arabian carbonate platform. Structurally, the area is bounded to the west by the Lekhwair High and to the east by the Sunaynah Trough, with a gentle regional dip trending toward the southeast. The stratigraphic sequence is dominated by thick Cretaceous carbonate successions deposited during multiple transgressive–regressive cycles, producing a range of depositional facies and reservoir qualities (Mufargi et al., 2025).

The primary intervals of interest—Shammar, Natih, and Shuaiba—form a vertically stacked carbonate sequence, each deposited under differing energy regimes and each presenting unique development and stimulation considerations. As illustrated in Figure 1, the Natih Formation is a laterally extensive, moderately thick carbonate that forms the main producing interval in the field. Directly beneath it lies the chalky and relatively homogeneous Shuaiba Formation, while the thin, dense Shammar unit occurs stratigraphically above the Natih. The cross-section also highlights the variability in thickness and facies distribution of these reservoirs across the structural trend from the Lekhwair High, through Daleel, to the Sunaynah Trough (Al Badi et al., 2025).

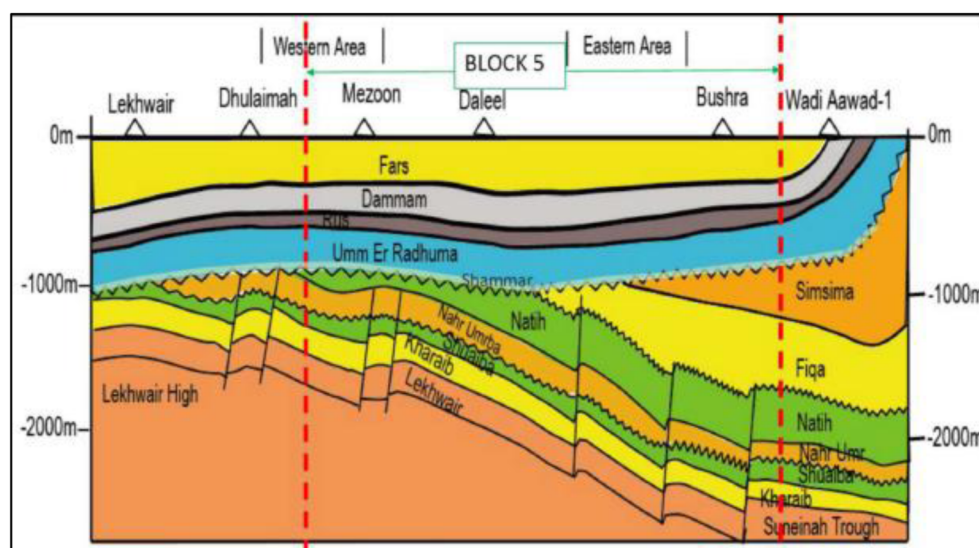


Figure 1—Regional geological cross-section across Block 5 highlighting the stratigraphic positioning of the Natih, Shuaiba, and Shammar (Al Badi et al., 2025).

Primary Reservoir Targets in Recent Campaign

After evaluating the production performance from previous campaigns, the Natih and Shuaiba formations were identified as the most promising targets for Large-Volume Acidizing (LVA) in Block 5. The Natih has consistently delivered significant oil gains, while the Shuaiba has shown measurable, though typically more moderate, improvements. This is illustrated in Figure 2, which presents an example of oil production uplift following LVA treatments in these formations.

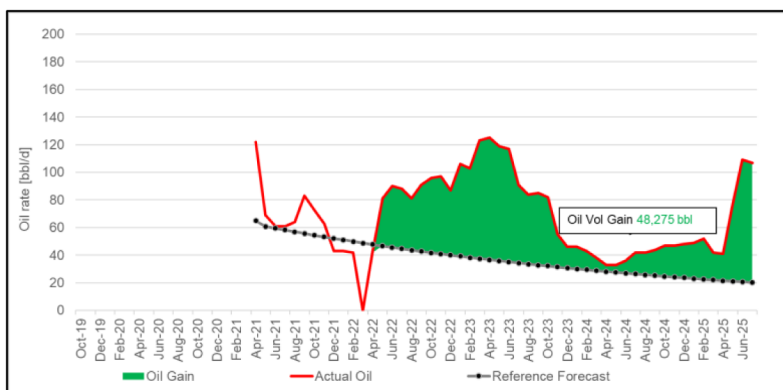


Figure 2—Example of production response from Well-14 in the Natih Formation following Large-Volume Acidizing (LVA). The plot compares actual post-treatment oil rates (red) against the reference forecast (black dotted line), highlighting an estimated cumulative oil gain of 48,275 bbl (green shaded area).

Based on these results, the 2025 stimulation program has been designed to focus exclusively on Natih and Shuaiba wells, excluding Shammar from the current campaign. In the Natih, cyclic carbonates with alternating porous and tight intervals require precise diversion strategies to ensure full lateral coverage. In the Shuaiba, its relatively uniform lithology but very tight matrix necessitates the use of deep-penetrating retarded acid systems—such as the newly deployed NG-SPRA—to initiate effective wormhole development. This targeted approach leverages formation-specific stimulation designs to maximize oil recovery potential and optimize return on investment, building on insights gained from previous LVA operations in Block 5.

The two principal formations targeted in Block 5—Natih and Shuaiba—exhibit notable differences in thickness, depositional setting, and reservoir quality, necessitating tailored stimulation approaches for each interval, as summarized in Table 1.

Table 1—Summary of reservoir properties and stimulation challenges for Natih, Shuaiba, and Shammar Formations in Block 5 (Mufargi et al., 2025).

Attribute	Natih	Shuaiba
Well Share (%)	~55%	~40%
Stratigraphic Position	Middle	Deepest
Reservoir Units	C3b, C2b	D or B1
Lithology	Heterogeneous carbonate (tight and porous streaks)	Chalky limestone with homogeneous texture
Reservoir Thickness	Moderate, multilayered	Typically single-layer
Porosity & Permeability	Variable porosity; permeability <1 mD	Tight matrix; low permeability
Completion Style	OH, single or dual lateral	OH, single lateral
Lateral Length (m)	569–1425	339–1209
Bottomhole Temperature	~69 °C (157 °F)	~77 °C (170 °F)
Bottomhole Pressure (psi)	1000–2000	600–1500
Gross Production Rate	Up to 200 bpd	Up to 150 bpd
Water Cut	Up to 50%	Up to 50%
Stimulation Challenges	Multilayer zones require diversion and acid tapering	Chalky, tight rock requires deeper penetration

Natih Formation – The most developed and widely produced reservoir in Block 5, consisting of cyclic carbonates with alternating porous and tight layers. Its complex stratigraphy and variable permeability demand careful acid placement and diversion control to ensure uniform stimulation along horizontal sections.

Shuaiba Formation – Deposited in lower-energy platform settings, this interval lies directly beneath the Natih and is composed of chalky, low-permeability carbonates. While its tight matrix limits natural productivity, its relative lithological uniformity makes it responsive to deep-penetrating acid systems, particularly when combined with foamed diversion.

Advancements in Acid Stimulation Technology

The Large-Volume Acidizing (LVA) program in Block 5 has progressed from a pilot concept to a standard stimulation method through continuous improvements in acid chemistry, diversion design, and operational execution. Over time, refinements in fluid systems, placement strategies, and treatment sequencing have improved coverage and production outcomes.

In 2025, the program advanced further with the introduction of the New Generation Single-Phase Retarded Acid (NG-SPRA) for deeper penetration and reduced dosage, alongside the first deployment of High-Frequency Pressure Monitoring (HFPM) to assess fluid entry and diversion performance in real time. [Figure 3](#) summarizes the evolution of LVA in Daleel, culminating in the 2025 integration of NG-SPRA and HFPM as key advancements.

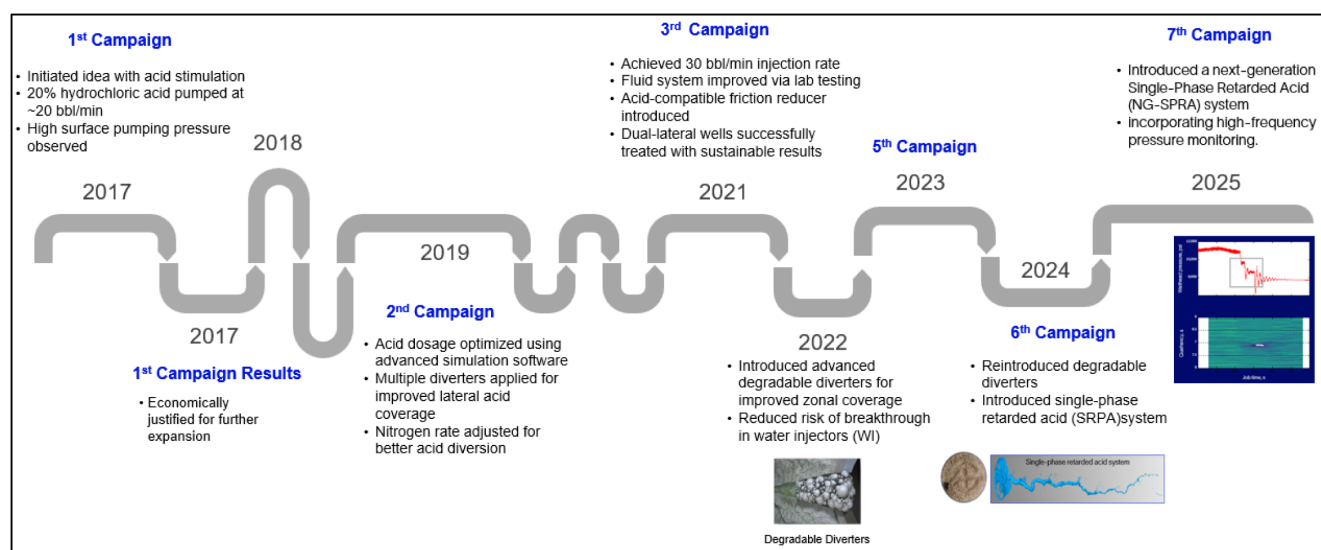


Figure 3—Chronological evolution of the Large-Volume Acidizing (LVA) stimulation campaigns in Block 5 from 2017 to 2025

Evolution of the Large-Volume Acidizing (LVA) Strategy in Block 5

The Large-Volume Acidizing (LVA) program in Block 5 began in 2017 with 20% HCl, surface-bullheaded at ~20 bbl/min, targeting long open-hole horizontal completions without coiled tubing (CT) or multistage isolation. Despite limited placement control and high treating pressures, the initial production uplift proved the concept's technical and economic viability ([Mufargi et al., 2025](#)).

- From 2019–2021, the program shifted toward enhanced diversion and nitrogen-energized foamed acid systems. Chemical diverters were introduced to mitigate heel-dominated entry, while foam improved diversion efficiency, maintained live acid delivery without CT, and aided post-stimulation cleanup—advantages well suited to Block 5's tight, heterogeneous carbonates, consistent with prior studies ([Mufargi et al., 2025](#)).

- By 2022–2023, engineered acid systems and degradable diverters were incorporated to improve lateral coverage in more heterogeneous wells. Retarded acid systems extended matrix penetration, while degradable diverters with controlled dissolution temporarily sealed dominant flow paths, leading to more uniform placement and sustained production gains (Mufargi et al., 2025).
- In 2024, Single-Phase Retarded Acid (SPRA)—a clear HCl-based fluid with water-soluble retarders—was deployed. Laboratory results confirmed deeper wormhole penetration, reduced PVBT, and improved acid efficiency compared with conventional HCl (Daeffler et al., 2021). In Block 5, combining SPRA with degradable diverters enhanced stimulation coverage toward the toe, reduced acid volumes, and retained the operational simplicity of CT-free bullheading (Mufargi et al., 2025).

Latest New Technology Implementation

New Generation Single-Phase Retarded Acid (NG-SPRA) Technology. Recent developments in acidizing technology have led to the introduction of a next-generation Single-Phase Retarded Acid (NG-SPRA) system, designed to deliver improved wormhole penetration, thermal stability, and operational efficiency in carbonate stimulation. Unlike conventional hydrochloric acid (HCl) or emulsified acid systems, NG-SPRA achieves controlled reaction rates while maintaining low viscosity, enabling high-rate pumping through long open-hole horizontals without exceeding wellhead pressure limitations (Danican et al., 2024).

The fluid can be formulated to match the carbonate dissolution capacity of up to 28 wt% HCl, with retarding agents that are thermally stable up to 325°F and polymer-free. Figure 4 presents static dissolution test results comparing NG-SPRA with unmodified HCl at various concentrations and temperatures, demonstrating significant retardation capability, particularly under high-temperature conditions.

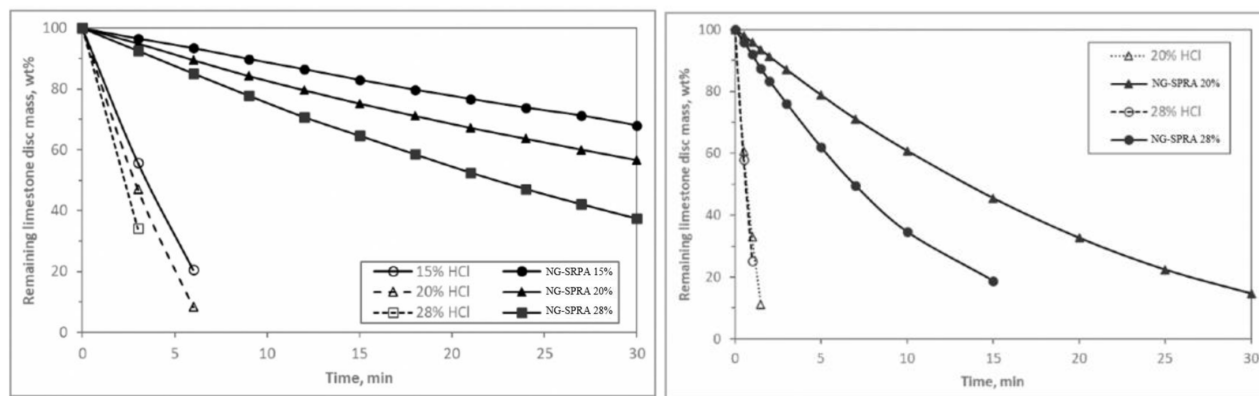


Figure 4—Static dissolution rate comparison between NG-SPRA and unmodified HCl at varying concentrations and temperatures—left: ambient temperature; right: 140 °F (adapted from SPE-217841-MS).

Laboratory evaluations—including static dissolution tests, thermal aging, acid strength verification, and matrix core flow experiments—demonstrated superior retardation performance and deeper wormhole formation compared to unmodified HCl and emulsified acids. In Figure 5, wormhole visualizations from Indiana limestone core tests clearly show the more uniform and extended penetration achieved with NG-SPRA, while pore volume to breakthrough (PVBT) results confirm its higher efficiency at multiple injection.

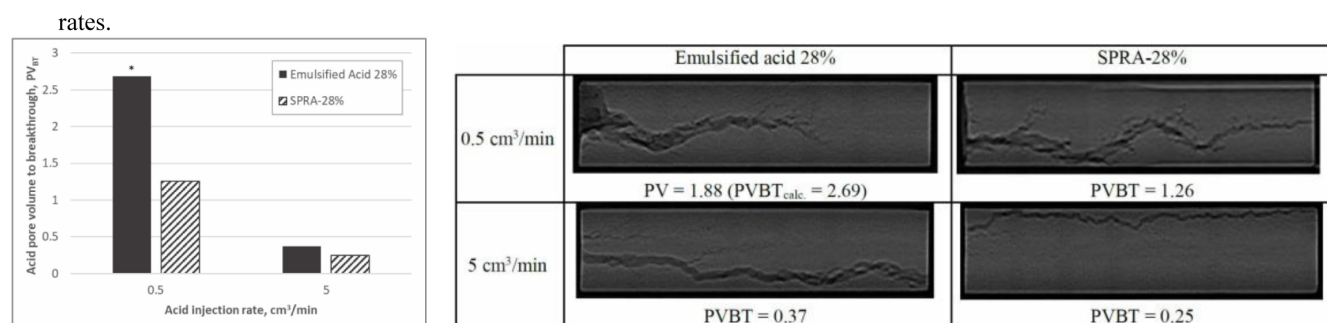


Figure 5—Wormhole visualization and PVBT comparison for NG-SPRA vs. conventional acid (adapted from SPE-217841-MS).

In Block 5 operations, the deployment of the Next-Generation Single-Phase Retarded Acid (NG-SPRA) in the 2025 LVA campaign represents a major advancement in stimulation fluid technology. Unlike conventional HCl or earlier retarded acid systems, NG-SPRA is formulated to maintain a slower, more controlled reaction rate with the carbonate matrix, allowing acid to penetrate deeper into the formation before significant spending occurs. This extended penetration improves stimulation coverage in heterogeneous carbonate intervals, particularly in low-permeability zones that are often bypassed during high leak-off treatments.

By reducing near-wellbore reaction losses and eliminating the need for emulsifiers or complex mixing processes, NG-SPRA also simplifies field operations while preserving the high-rate bullheading capability essential for long open-hole laterals in Block 5. Laboratory evaluations and early field results indicate that NG-SPRA can deliver longer wormhole structures with lower pore volume to breakthrough (PVBT), translating to improved acid utilization and more effective reservoir contact. This improvement directly addresses the uneven stimulation coverage observed in previous campaigns and provides a scalable, cost-effective solution for enhancing production in tight carbonate reservoirs.

Deployment of High-Frequency Pressure Monitoring (HFPM). High-Frequency Pressure Monitoring (HFPM) is a nonintrusive diagnostic method that determines reservoir fluid-entry points by acquiring high-frequency surface pressure data during carefully coordinated pump shut-downs. Through cepstrum-based analysis of water-hammer (WH) transients, tube-wave reflection times are extracted and mapped—with associated uncertainty bounds—to probable fluid-entry locations along the lateral, enabling near-real-time evaluation of zone participation and diversion effectiveness without the need for downhole instrumentation (Estrada et al., 2022; Zeng et al., 2023).

The operational workflow follows an iterative observe–diagnose–divert–re-diagnose loop: after pumping a treatment stage, the rate is reduced to a low, stable value for 2–3 minutes, followed by a simultaneous shut-down to capture a clean transient. The resulting diagnostic identifies active entry points, after which a diverter pill may be deployed if redistribution is required. Pumping is then resumed—briefly ramping to ~50 % of the design rate to initiate new breakdown—before repeating the shut-down and diagnosis. This cycle continues until no further improvement in lateral coverage is observed. Figure 6 (adapted from Zeng et al., 2023) depicts the full HFPM-guided stimulation loop, illustrating the sequential steps that enable targeted and efficient fluid placement.

To generate high-quality water-hammer data, specific shut-down practices are recommended. A controlled step-down from the main rate to approximately 10–20 bbl/min for 2–3 minutes before a simultaneous stop improves transient clarity. After diversion, a brief rate increase (approximately 50% of design) before the next shut-down helps ensure new breakdown prior to diagnosis (Zeng et al., 2023). The algorithmic foundation of HFPM reconciles cepstrum-derived reflection times with plausible tube-wave velocities and known interval depths, updating entry-depth probabilities as additional shut-downs are

processed. Where velocity uncertainty is significant, a calibration step using a high-viscosity spacer (viscous pill) can improve depth resolution (Estrada et al., 2022).

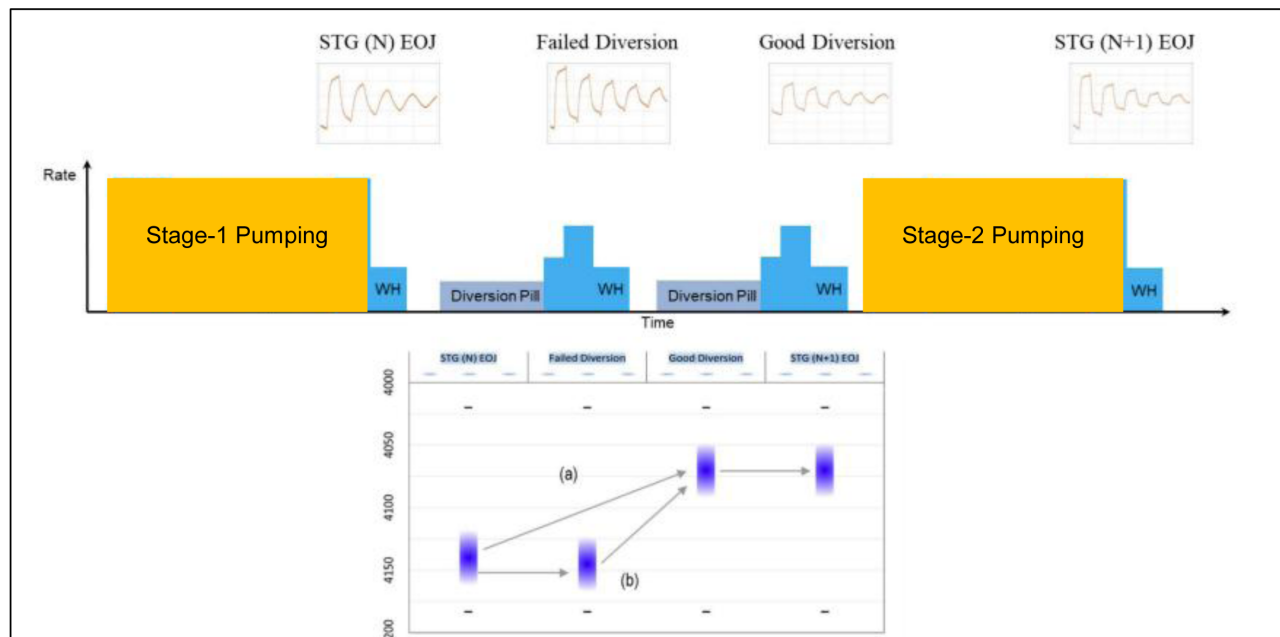


Figure 6—Workflow of HFPM-guided stimulation showing observe–diagnose–divert–re-diagnose sequence (adapted from Zeng et al., 2023 (SPE-214867-MS))

An example from Estrada et al. (2022), shown in Figure 7, presents a typical water-hammer event. The upper panel depicts the recorded pressure transient during shut-down, and the lower panel displays the corresponding cepstrum output with identified reflection-time picks. Field deployments have shown strong agreement between HFPM-diagnosed entry points and independent measurements such as tracers, downhole video, and surface seismic, while offering a small operational footprint, low cost, and rapid turnaround. Known limitations include dependence on clean, synchronized shut-downs, reduced sensitivity when a dominant heel conduit masks weaker reflections, and the inability to directly characterize fracture dimensions (Estrada et al., 2022; Zeng et al., 2023).

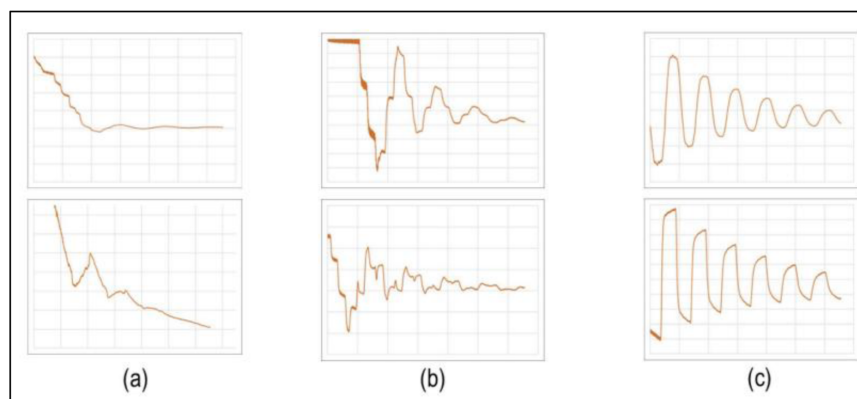


Figure 7—Typical water hammer wave signals: (a) Weak or quickly declining water hammer. (b) Water hammer wave with strong noises due to nonsimultaneous pump shutdown. (c) Clear water hammer wave generated from recommended practice (Estrada et al., 2022)

In situations where a distinct water-hammer transient cannot be observed during the standard shut-down diagnostic—owing to factors such as low signal strength, depleted reservoir conditions, or signal damping

in long open-hole completions—an alternative rate-changing approach can be applied prior to the main treatment shut-in. This technique consists of inducing short-duration, controlled pump-rate fluctuations to generate a consistent sequence of pressure oscillations in the wellbore. The resulting pulsations can enhance tube-wave excitation and improve the detectability of reflection patterns in the HFPM cepstrum domain.

As illustrated in Figure 8, the rate changing method produces a repeatable cepstrum response from which a consistent reflection-time signal can be identified, even in the absence of a strong single water-hammer event. In this example, the top panel shows the cepstrum output highlighting stable reflection-time bands, while the bottom panel presents the corresponding surface pressure trace during the modulation sequence. This approach provides an effective contingency for HFPM diagnostics, enabling reliable entry-depth estimation when conventional shut-down transients are insufficiently distinct.

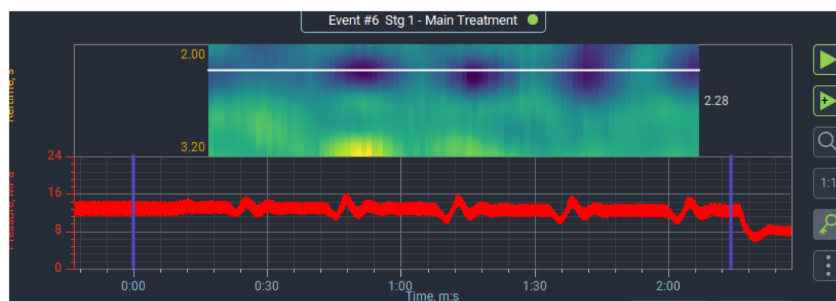


Figure 8—Example of rate-changed-induced pressure modulation as an alternative HFPM diagnostic method. The top panel shows the cepstrum output with stable reflection-time bands, while the bottom panel displays the corresponding surface pressure trace during the modulation sequence, demonstrating effective entry-depth estimation in the absence of a strong single water-hammer event.

HFPM Pilot for Acid Placement Diagnostics in Block 5. In Block 5, well surveillance capabilities—particularly for producing wells—are limited. The absence of downhole monitoring tools such as permanent pressure/temperature gauges, production logging systems, or fiber-optic distributed sensing presents a significant challenge for assessing stimulation effectiveness. This constraint is especially pronounced in long open-hole horizontal completions, where heterogeneity in injectivity, reservoir depletion, and uncertain diversion efficiency can lead to uneven stimulation coverage. Without direct measurement of fluid placement, post-treatment evaluation has historically relied on indirect indicators such as production response trends, which are influenced by multiple reservoir and operational variables and therefore cannot isolate stimulation performance with high confidence.

The introduction of HFPM offers a cost-effective, nonintrusive solution to this diagnostic gap. By utilizing only high-frequency surface pressure and synchronized rate/volume data, HFPM can infer fluid-entry distribution in near real time—without the need for additional hardware installation or intervention. This capability is particularly valuable in Daleel's context, where both capital and operational constraints limit the feasibility of mechanical isolation, tracer injection, or coiled-tubing-based diagnostics for most LVA campaigns.

The first planned HFPM deployment in Daleel will be executed in observation mode during a long open-hole Large-Volume Acidizing (LVA) pilot treatment. The well will be stimulated using a combination of foamed diversion and degradable diverters to maximize lateral coverage. HFPM diagnostics will be strategically scheduled:

- After the baseline acid slug(s) to establish an initial fluid-entry profile.
- Following each diverter stage to evaluate changes in entry distribution.
- Selectively toward the end of the treatment to assess final coverage.

The pilot program will deliver three key diagnostic outputs:

- Time-lapse entry-depth probability profiles comparing pre- and post-diversion fluid-entry distributions.
- Diversion-success metrics defined as the fraction of shut-down events showing statistically significant shifts in entry-point distribution.
- Lateral coverage indicators quantifying the proportion of the open-hole section inferred to have accepted fluid at least once during the treatment.

By providing quantitative placement diagnostics in an environment where no equivalent surveillance data currently exists, HFPM will allow the Daleel stimulation team to calibrate design parameters—including acid stage volumes, diverter types/dosages, and pumping schedules—based on actual in-situ performance rather than purely on modeled predictions. Lessons learned from this pilot will directly inform candidate selection criteria, stimulation sequencing, and real-time execution strategies in future campaigns.

In future deployments, HFPM will be integrated into a closed-loop execution model, where diagnostic feedback will guide dynamic adjustments to pumping schedules, diversion sequencing, and rate transitions during the treatment itself. This capability has the potential to significantly improve stage coverage, enhance stimulation efficiency, and reduce the variability of treatment outcomes across the heterogeneous carbonate formations of Block 5. Ultimately, HFPM could become a standard component of LVA operations in Daleel, bridging the current surveillance gap and enabling data-driven optimization of stimulation practices (Zeng et al., 2023; Estrada et al., 2022).

Methodology and Treatment Design Framework

Candidate Selection Criteria and Workflow for the 2025 Block 5 LVA Campaign

Candidate selection for Large Volume Acidizing (LVA) in Block 5 has evolved into a structured, data-driven process that blends quantitative analysis with multidisciplinary review. This methodology ensures that stimulation resources are allocated to wells with the highest likelihood of delivering sustainable incremental production or injection gains, while managing operational risks (Al-Mufargi et al., 2025).

The process is cyclical and integrates three key stages:

- **Comprehensive Data Gathering and Quality Control** – Reservoir and geological properties (porosity, permeability, net-to-gross ratio), well configuration (lateral length, completion type), production/injection performance, pressure depletion indicators, and prior stimulation history are compiled and normalized to a consistent dataset.
- **Quantitative Scoring and Ranking** – Each parameter is assigned a weighting based on its historical correlation with stimulation success. The result is a composite score normalized to 100%, allowing direct comparison between candidates.
- **Peer Review and Operational Screening** – A cross-functional team of geoscientists, reservoir engineers, production technologists, and stimulation specialists reviews the ranked list, applying exclusion criteria (e.g., mechanical constraints, proximity to oil–water contact, breakthrough risks) and confirming operational feasibility.

Lessons learned from each campaign are fed back into the model, refining the weighting scheme and threshold cut-offs for the next year. Separate scoring matrices are applied for oil producers and water injectors to reflect differences in stimulation objectives and operational risks. The 2025 weighting scheme was tailored to maximize technical success as shown in Table 2:

- **Oil producers** – Highest weightings were assigned to normalized production per 100 m of net lateral length (12.5%), net-to-gross ratio (10%), permeability (10%), water-cut penalty (10%), and

production-decline penalty (10%). Additional parameters included open-hole length, porosity, oil saturation, proximity to OWC, previous LVA history, and reservoir pressure status.

- **Water injectors** – Key drivers included normalized injection rate (10%), net-to-gross ratio (10%), permeability (10%), breakthrough risk (10%), and open-hole length (8%), supported by porosity, decline rate, depletion, fault proximity, and previous LVA history.

Table 2—Candidate selection weighted scoring criteria

Parameter for Producer	Weight (%)	Parameter for Injector	Weight (%)
Total Open-Hole Length	5	Total Open-Hole Length	8
Reservoir Thickness (TVD)	2.5	Reservoir Thickness (TVD)	2.5
Faulting Proximity	5	Faulting Proximity	5
Net-to-Gross (NTG)	10	Net-to-Gross (NTG)	10
Average Porosity	5	Average Porosity	10
Average Oil Saturation	5	Average Oil Saturation	2.5
Normalized Production**	12.5	Normalized Injection**	10
Average Permeability	10	Average Permeability	10
Proximity to OWC (lower is better)	7.5	Water Breakthrough Risk	10
Previous LVA History	11	Previous LVA History	5
Average Water Cut (lower is better)	10	Injection Rate Decline	7.5
Rate Decline from Initial Rate	10	Reservoir Pressure Depletion	6
Reservoir Pressure Depletion	7		

This weighting distribution reflects field experience: for producers, uplift correlates strongly with rock quality and pre-job productivity; for injectors, success is linked to injectivity potential and managing connectivity to offset producers. Penalties for high water cut, OWC proximity, and breakthrough history help balance stimulation potential with flood management discipline (Al-Mufargi et al., 2025).

2025 Selected Candidates. Applying the scoring and review process yielded the following high-priority wells for the 2025 campaign:

- **Natih Producers**
 - **Well-P-1** – 61% Favorable geological setting; crosses multiple faults with no evidence of conductive leakage.
 - **Well-P-2** – 52% Lower gross production but operationally viable; selected for potential uplift.
- **Upper Shuaiba Producers**
 - **Well-P-4** – 61% Low normalized production with moderate water cut; good candidate for matrix acidizing.
 - **Well-P-3** – 56% Underperforming lateral targeted for coverage improvement.
 - **Well-P-5** - 49% Low NTG and high water cut; retained as a trial case for learning in Shuaiba stimulation.
- **Water Injectors**
 - **Well-I-1** – 80% (Rank 3): Strong injectivity potential; prioritized for capacity restoration.

- **Well-I-3** – 78% Supports two producers; observed breakthrough from another injector; design focused on containment.
- **Well-I-2** – 68% Strong early-time injectivity, suggesting high recovery potential post-stimulation.

Final selection included a mix of top-ranked candidates and lower-ranked "learning wells" to expand the operational dataset into different reservoir and completion settings.

Treatment Design Framework

For 2025, stimulation design continues to be rig-less, surface-bullhead LVA with chemical diversion—avoiding coiled tubing to reduce cost and operational complexity. Pre-job designs are simulation-led, incorporating reservoir heterogeneity, heel-bias risk, and predicted leak-off to optimize fluid placement (Al-Mufargi et al., 2025).

Figure 9 presents the comparison of open-hole (OH) lateral length and acid dosage per unit length for selected Natih and Upper Shuaiba (U. Shuaiba) wells in Block 5, comprising both producers (P) and injectors (I). The OH length is plotted on the primary (left) y-axis in meters, while the total acid dosage per length is shown on the secondary (right) y-axis in gallons per foot (gal/ft).

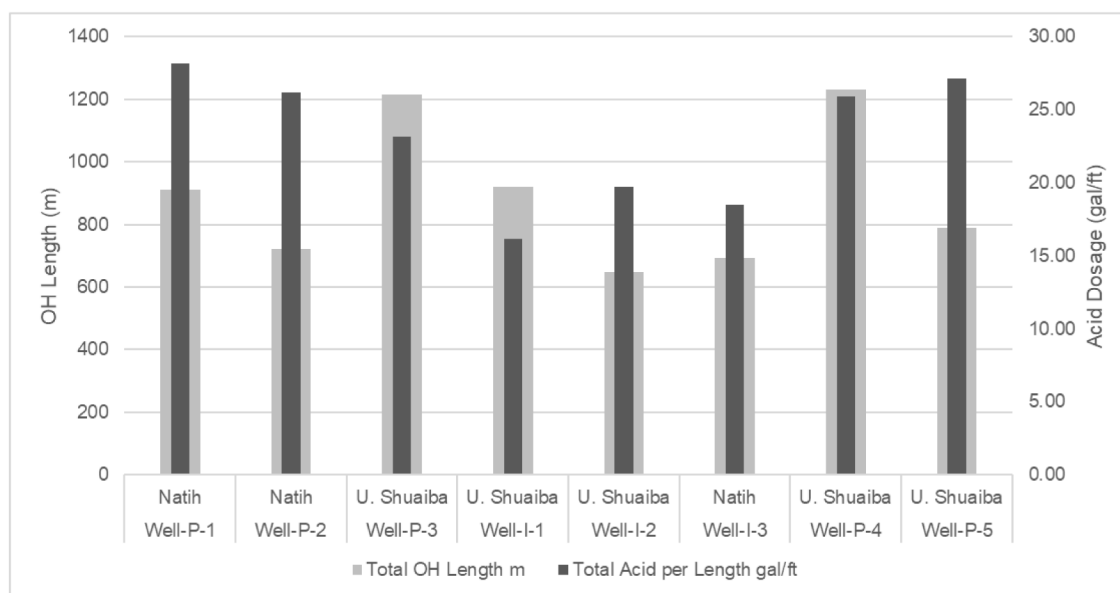


Figure 9—Open-hole (OH) lateral length and total acid dosage per unit length for Natih and U. Shuaiba producers and injectors in Block 5.

In the Natih Formation, Well-P-1 and Well-P-2 exhibit the longest laterals among all candidates, with OH lengths exceeding 1,200 m. These wells also received some of the highest acid dosages, approaching 27 gal/ft, reflecting the strategy to maintain deep acid penetration in extended laterals to offset potential near-heel fluid dominance. Well-I-3, the only Natih injector in this dataset, has a shorter lateral length (~900 m) and received a comparatively moderate dosage (~17 gal/ft), consistent with the reduced acid design approach for injectors to control fracture growth and minimize unwanted fluid communication with producers.

In the Upper Shuaiba Formation, OH lengths show greater variability. Well-P-3 and Well-P-4 are relatively long laterals (~1,100–1,250 m) and received higher acid dosages (~24–26 gal/ft), a deliberate choice to overcome the chalky, low-permeability matrix and enhance wormhole development. Well-P-5 also shows a similar dosage (~25 gal/ft) despite a slightly shorter lateral length (~1,050 m), indicating an emphasis on achieving comparable stimulation intensity across production wells. Conversely, injector wells (Well-I-1 and Well-I-2) in the Shuaiba Formation received markedly lower dosages (~15–17 gal/ft) and

have shorter laterals (<900 m), in line with the injector-specific design modifications implemented after the 2023 breakthrough events.

Overall, the data in Figure 9 demonstrate a consistent design philosophy:

- **Producer laterals** in both Natih and Shuaiba formations are treated with higher acid volumes (~26 gal/ft) per foot to ensure full-length stimulation and deep penetration.
- **Injector wells**, regardless of formation, receive shorter OH lengths and deliberately reduced acid dosages to maintain zonal control, minimize over-penetration, and reduce cross-flow risks.

This treatment allocation reflects the operational balance between maximizing productivity in producers and maintaining injection sustainability in waterflood management.

A comparison of the 2024 and 2025 oil producer candidates in Block 5 is shown in Figure 10, illustrating the open-hole (OH) lateral length (left y-axis) and the corresponding acid dosage per unit length (right y-axis). The 2024 candidate featured an OH length of approximately 1,070 m and was treated at a dosage of 27 gal/ft, designed to ensure deep wormhole penetration using single phase retarded acid (SPRA) and foamed diversion.

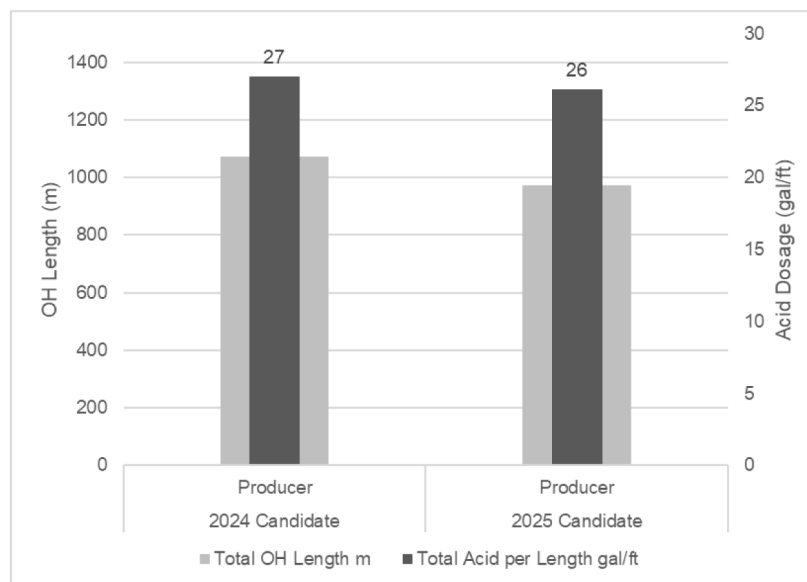


Figure 10—2024 vs 2025 LVA treatment design comparison.

In contrast, the 2025 candidate, while having a slightly shorter OH length (~970 m), was assigned a lower dosage of 26 gal/ft. This reduction reflects the adoption of the next-generation Single-Phase Retarded Acid (NG-SPRA) system, which offers slower reaction kinetics, improved wormhole efficiency, and reduced near-wellbore reaction losses. Laboratory and field trials have demonstrated that NG-SPRA can achieve comparable or greater penetration depths at lower volumes than conventional acid systems, thereby allowing optimization of acid loading without compromising stimulation reach.

Fluid Systems and Diversion Strategy.

- **Base fluid** – Nitrogen-energized foamed acid is utilized to enhance diversion efficiency and improve fluid placement along long open-hole laterals, leveraging its ability to increase apparent viscosity, reduce leak-off, and promote uniform coverage (Rahim et al., 2018).
- **Advanced acid system** – The 2025 campaign incorporates the Next-Generation Single-Phase Retarded Acid (NG-SPRA), engineered to deliver deeper penetration into tight carbonate matrices by reducing near-wellbore reaction rates and improving acid utilization. This formulation is

expected to translate into greater stimulation effectiveness and improved post-treatment production performance.

- **Diversion strategy** – Multisize degradable diverters are deployed between acid stages to temporarily restrict dominant flow channels, enabling redirection of treatment fluids into previously bypassed or lower-permeability intervals, thereby enhancing overall lateral coverage.

Injector-Specific Adjustments. In Block 5, operational experience from the 2023 LVA campaign revealed that conventional acid stimulation of injector wells could trigger rapid fluid communication with adjacent producers. To address this, the 2024/2025 program adopted a tailored injector stimulation design focused on controlled acid placement and zonal management (Al-Mufargi et al., 2025). Key measures included:

- **Step-rate test-driven pumping** to establish formation parting pressure and ensure acid injection remained within a controlled matrix regime until acid-rock reaction was observed.
- **Reduced acid dosage** (~18 gal/ft vs. ~26 gal/ft for producers) to limit penetration depth and mitigate breakthrough risk.
- **Use of degradable diverters** to temporarily block dominant flow paths, promoting more uniform acid distribution across all target intervals.

This targeted approach improved placement control, reduced cross-flow risk, and supported more sustainable waterflood performance.

Treatment Design Simulation

The stimulation design for Block 5 leveraged a simulation-led workflow in which all fluids were delivered via surface bullheading through the tubing, eliminating the need for coiled tubing. Pre-job modeling was used to predict zonal acid placement, wormhole propagation, and the impact of alternating acid and diverter stages on stimulating distal lateral sections. These rainbow-colored pre-treatment profiles illustrated expected wormhole extent and skin reduction, while post-treatment re-simulation—represented by black traces—incorporated actual job data to refine the estimates of achieved stimulation. Although field results have shown productivity gains consistent with these predictions, it is important to note that no direct diagnostic calibration (e.g., tracers, image logs, or pressure pulse analysis) has yet been applied; future campaigns should incorporate such tools to validate and enhance simulation accuracy.

To ground these simulations in physics-based data, the NG-SPRA laboratory-to-field modeling approach described by Ali and Ziauddin (2020) in SPE-217841-MS was embedded into the design process. More than 100 linear core-flow experiments on Indiana limestone and Silurian dolomite were conducted to calibrate a PVBT model for NG-SPRA fluids. The mechanistic wormholing model was then used to scale results to radial, field-scale acid-efficiency curves. At 325 °F and 1 gal/ft/min, the acid volume required to achieve a 4-ft wormhole was shown to be approximately sixty times lower for SPRA-28% than for unmodified 28% HCl (Ali and Ziauddin 2020). This substantial improvement in acid efficiency directly informed rate selection, dosage, and acid strength in the Block 5 designs, ensuring deeper wormhole penetration and better acid utilization in tight carbonate intervals.

The simulation outcomes varied across formations. In the Natih Formation, a dense and continuous wormhole network extended from heel to toe, with skin values consistently below -2.5 in thief zones, confirming strong stimulation efficiency even in long laterals. This behavior is exemplified by Well-P-1 (Figure 11), where the simulation predicted and achieved an average skin of -2.86 and demonstrated uniform fluid coverage along the entire lateral length.

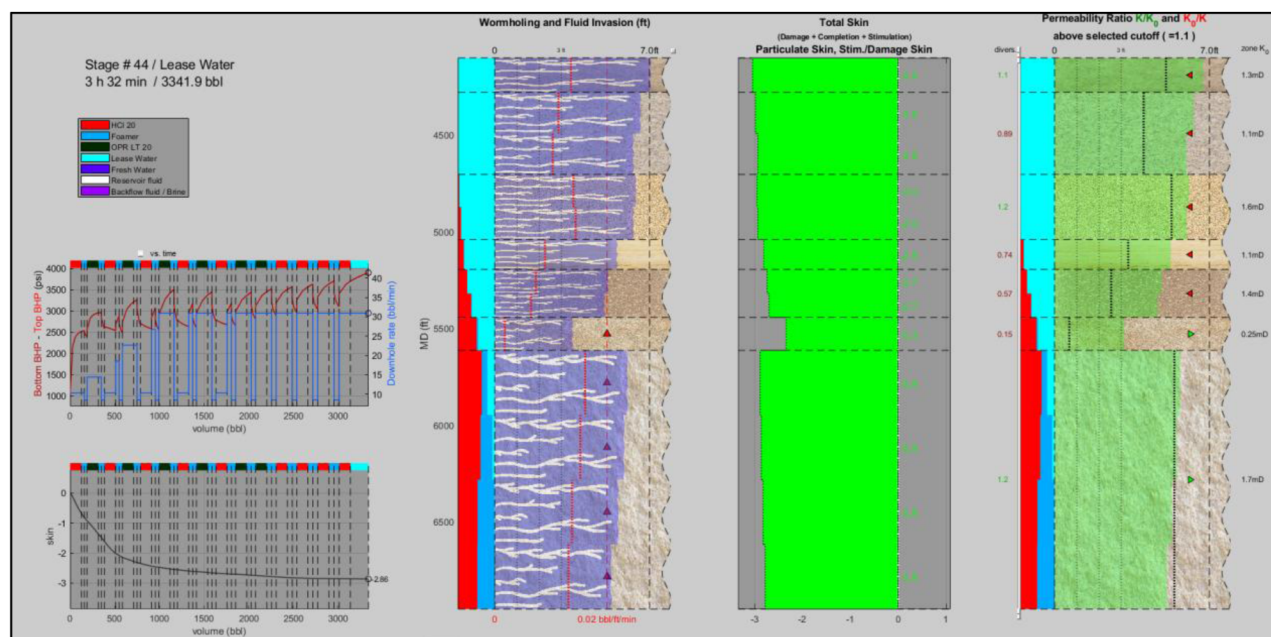


Figure 11—Pre- and post-treatment stimulation simulation results for Well-P-1 in the Natih Formation. The wormholing and fluid invasion panel (left) shows a continuous wormhole network from heel to toe, indicating effective acid coverage along the entire lateral. The total skin panel (center) reflects a significant reduction in near-wellbore skin, averaging -2.86 , while the permeability ratio panel (right) confirms uniform improvement in effective permeability across the treated interval.

In the Shuaiba Formation, stimulation effectiveness was influenced by reservoir heterogeneity. Overall, treatment was more pronounced in the heel section, while toward the toe, the simulated skin profile increased toward slightly higher values—suggesting reduced penetration depth. This trend is reflected in Well-P-3 (Figure 12), which achieved an average skin of -2.90 , with modeling indicating preferential acid placement in the heel section due to more favorable reservoir properties.

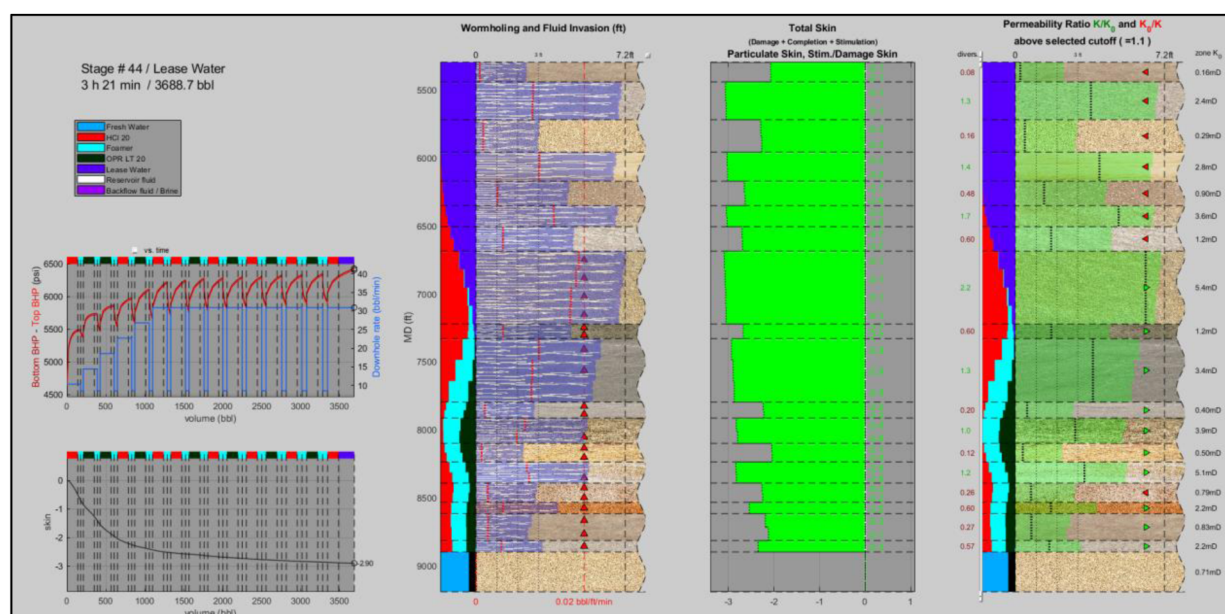


Figure 12—Pre- and post-treatment stimulation simulation results for Well-P-3 in the Shuaiba Formation. The wormholing and fluid invasion panel (left) highlights stronger acid penetration in the heel section, with reduced coverage toward the toe. The total skin panel (center) shows an average skin reduction to -2.90 , and the permeability ratio panel (right) indicates preferential improvement in the heel due to more favorable reservoir properties in that section.

These results confirm that simulation-guided, bullheaded LVA treatments—enhanced with NG-SPRA modeling inputs—can deliver full-length stimulation in the Natih and partially in the Shuaiba, while highlighting the importance of tailoring stimulation designs to the specific reservoir characteristics of each formation.

Further Validation Works and Future Improvements

Pre-job simulations in Block 5 are used to predict wormhole propagation and skin reduction, while post-job re-simulations incorporate actual execution data to refine these predictions. However, due to the absence of routine downhole surveillance, direct calibration remains limited—apart from planned PLT acquisition in injector wells and indirect validation through production performance.

For the 2025 campaign, High-Frequency Pressure Monitoring (HFPM) has been introduced as a pilot diagnostic to provide near-real-time fluid-entry evaluation during LVA operations. This will enable each treated well to have its stimulation simulation later updated with HFPM-derived observations, creating the first true closed-loop design–execution–feedback cycle for the field. Such integration is expected to enhance the accuracy of future modeling and provide a more reliable assessment of diversion effectiveness and stimulation coverage (Estrada et al., 2022; Zeng et al., 2023).

Large-Volume Acidizing (LVA) – Results and Discussion

Initial Production Performance Assessment of 2025 Candidate Wells

The initial post-stimulation results from the 2025 Large-Volume Acidizing (LVA) campaign demonstrate a marked improvement in well productivity with the deployment of the Next-Generation Single-Phase Retarded Acid (NG-SPRA) system compared to the 2024 campaign, which utilized the conventional Single-Phase Retarded Acid (SPRA). As summarized in Table 3 and Figure 13, the 2024 campaign—covering four Natih reservoir wells—recorded an average gross production increase from 62.75 bbl/d to 115.00 bbl/d, corresponding to a gross Fold of Increase (FOI) of 1.83, and an oil rate increase from 45.75 bbl/d to 64.25 bbl/d, yielding an oil FOI of 1.40. By contrast, the 2025 NG-SPRA campaign, implemented on two Natih wells, achieved a significantly higher average gross rate uplift from 33.27 bbl/d to 152.96 bbl/d (gross FOI = 4.60) and oil rate increase from 17.89 bbl/d to 73.27 bbl/d (oil FOI = 4.10). This translates to more than 2.5 times the gross productivity gain and almost three times the oil productivity gain compared to the SPRA program.

Table 3—Candidate selection weighted scoring criteria

Year	Acid System	Zone	# Wells	Avg Rate Before (bbl/d)	Avg Rate After (bbl/d)	FOI
2024	SPRA	Natih	4	62.75 (Gross) / 45.75 (Oil)	115.00 (Gross) / 64.25 (Oil)	1.83 (Gross) / 1.40 (Oil)
2025	NG-SPRA	Natih	2	33.27 (Gross) / 17.89 (Oil)	152.96 (Gross) / 73.27 (Oil)	4.60 (Gross) / 4.10 (Oil)

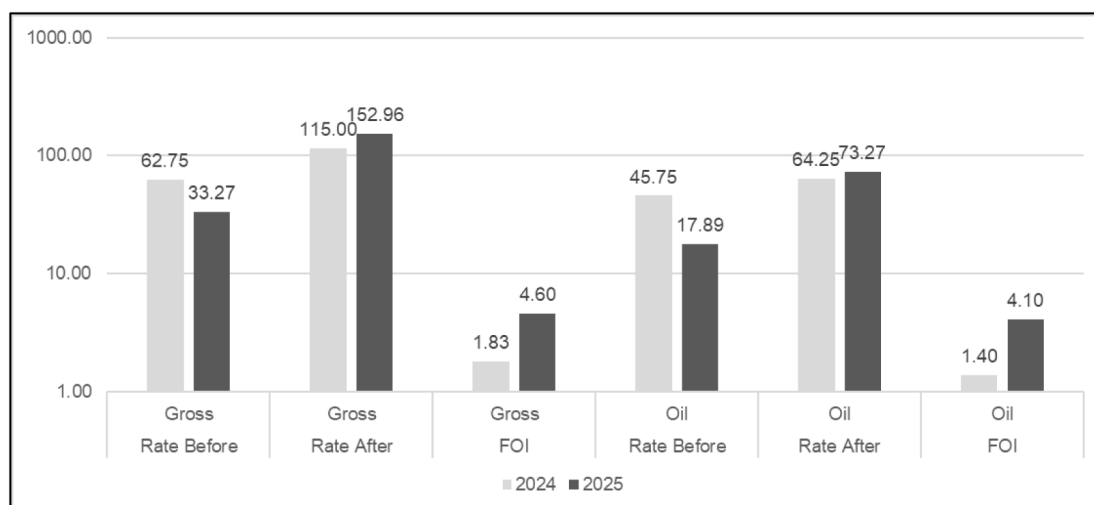


Figure 13—Comparison of average gross and oil production performance for the 2024 SPRA campaign (four Natih wells) and the 2025 NG-SPRA campaign (two Natih wells), showing pre- and post-stimulation rates and corresponding Fold of Increase (FOI). The 2025 NG-SPRA jobs achieved more than 2.5× higher gross rate improvement and nearly 3× higher oil rate improvement compared to the 2024 SPRA program.

The improved performance is attributed to NG-SPRA's optimized chemistry, which extends the acid–rock reaction time without the use of emulsified systems, enabling deeper wormhole propagation in the tight carbonate matrix of the Natih Formation. This extended reactivity reduces premature acid spending near the wellbore and facilitates more uniform coverage along long open-hole laterals. NG-SPRA maintains the low viscosity of conventional HCl, allowing high-rate bullheading with minimal friction losses, while improving acid penetration into lower-injectivity intervals. When preceded by nitrogen-energized foamed acid preflushes, the system further enhances diversion efficiency, mitigates heel-dominated placement, and promotes effective post-treatment cleanup by utilizing stored gas energy.

Limitations of the Preliminary Evaluation. While the 2025 NG-SPRA campaign shows strong potential for enhancing stimulation efficiency in heterogeneous carbonate reservoirs, several limitations mean the results should be viewed as indicative rather than conclusive. The dataset is small—two Natih wells versus four in the 2024 SPRA campaign—making it sensitive to well-specific factors. No normalization has been applied for differences in reservoir quality (kh), pressure, lateral length, or completion design, all of which can significantly affect FOI.

The evaluation is based only on short-term post-stimulation tests, capturing early clean-up and transient flow without confirming long-term performance or decline trends. Diagnostic surveillance was also limited, aside from planned PLT in injectors and indirect production analysis, leaving modeled stimulation coverage unverified. Future work should expand the dataset, normalize for reservoir and completion variables, incorporate longer-term production tracking, and integrate real-time diagnostics (e.g., HFPM, tracers, PLT) to better correlate placement with performance and validate NG-SPRA's true advantage in Block 5.

High-Frequency Pressure Monitoring (HFPM) Evaluation

High-Frequency Pressure Monitoring (HFPM) was deployed across all 2025 candidate wells to deliver real-time diagnostics on fluid placement and diversion effectiveness during stimulation. Although the full evaluation for all wells is still ongoing, Well P-1 has been selected as an initial case study to illustrate the HFPM methodology, the interpretation workflow, and the nature of insights this technique can provide (Figure 14).

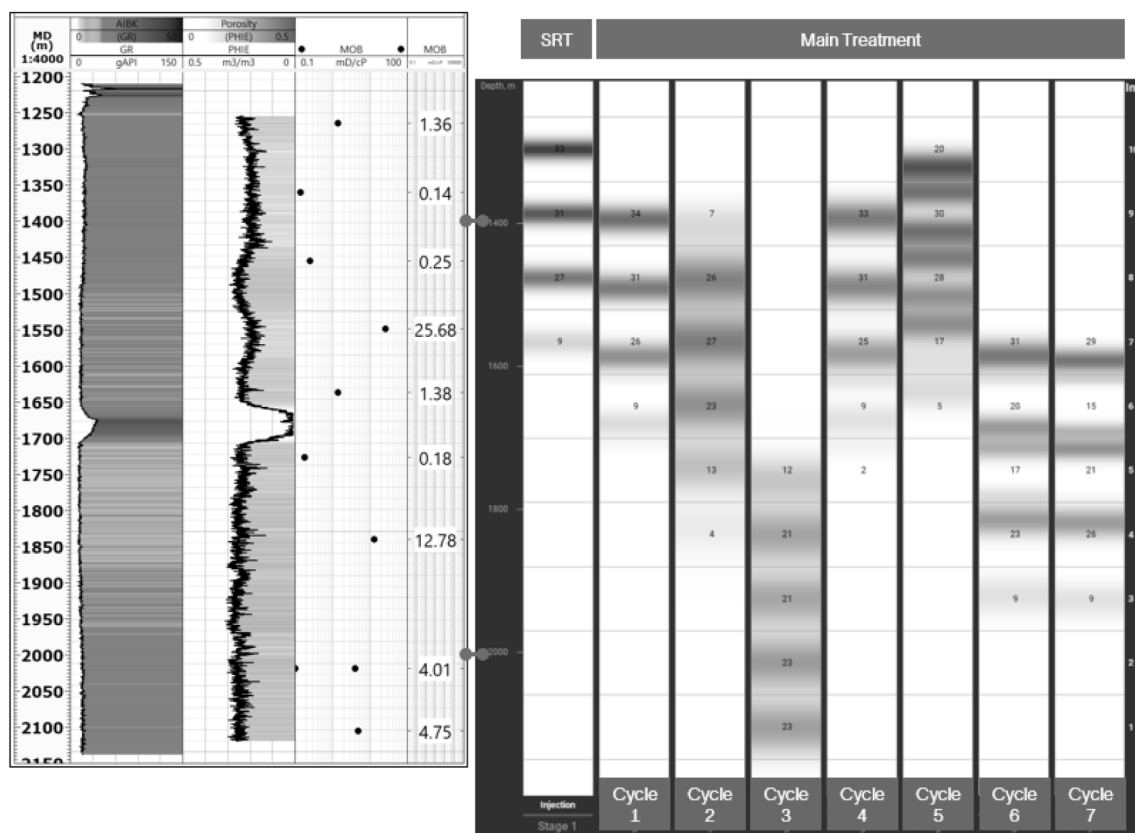


Figure 14—HFPM-based diversion analysis for Well-P-1, illustrating zone-by-zone fluid placement throughout the stimulation.

In this case, the HFPM analysis divided the reservoir section into three distinct zones—heel, mid, and toe—allowing a zone-by-zone interpretation of stimulation behavior. During the Step Rate Test (SRT), fluid entry was initially concentrated in the heel, as indicated by the pressure-hammering response. Once the main treatment began, with foamed diversion in place, stimulation shifted toward the mid-zone during the first shut-in cycle. Midway through the treatment, stimulation appeared to migrate toward both mid and toe sections, most likely influenced by a localized high-mobility interval at 12.78 mD·ft. As treatment progressed, foam stability concerns caused a temporary return of fluid placement toward the heel. In the final treatment cycles, however, the dominant stimulation response was again observed in the mid-zone, suggesting that this interval had the highest probability of receiving fluid placement during the later stages.

The interpretation utilized the HFPM probabilistic algorithm with a resolution of ± 80 ft. While HFPM has not yet been fully validated for foamed treatments, its integration with unlimited entry modeling and reservoir quality characterization provides a reasonable level of confidence in the conclusions.

The Well P-1 example demonstrates HFPM's capability to deliver time-lapse, depth-specific fluid placement data without disrupting live pumping operations. However, this represents only the initial phase of the pilot application. Due to technical constraints, HFPM data for the remaining 2025 campaign wells is still under processing. Future work will extend the diversion analysis to all stimulated wells and integrate the findings with production simulation results after 3–4 months of post-stimulation monitoring. These consolidated results will be presented in a separate, dedicated paper to evaluate the correlation between diversion effectiveness and sustained production performance.

Overall Stimulation Outcomes in Natih Formation

As of August 2025, Natih wells demonstrated the most consistent and substantial response to LVA, with an average FOI of ~ 2.7 —meaning post-stimulation oil rates averaged 2.7 times their pre-stimulation values. Figures 15–17 illustrate three representative Natih wells, each showing a sharp production surge

immediately after treatment, followed by a sustained plateau well above the pre-job decline trend. In [Figure 15](#), Well 10's oil rate increased from ~80 BOPD to nearly 150 BOPD immediately post-stimulation, maintaining >100 BOPD for months and delivering a cumulative oil gain of 98,781 bbl. [Figure 16](#) shows Well 14, which achieved a cumulative gain of 48,275 bbl, with production consistently above the forecast decline curve. Similarly, [Figure 17](#) highlights Well 23, where stimulation added over 56,000 bbl of cumulative oil.

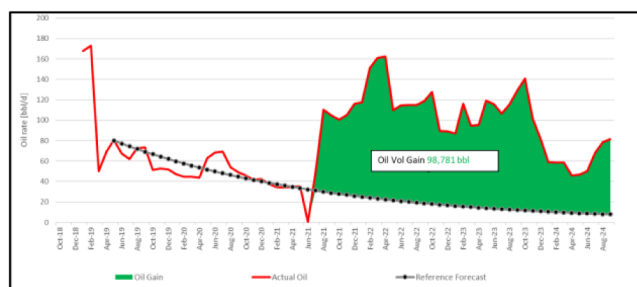


Figure 15—Post-stimulation production trends for Natih, Well 10.

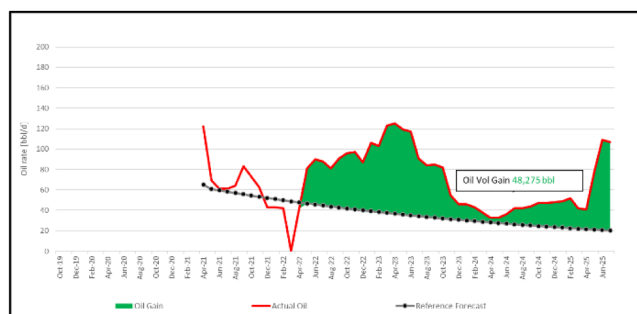


Figure 16—Post-stimulation production trends for Natih, Well 14.

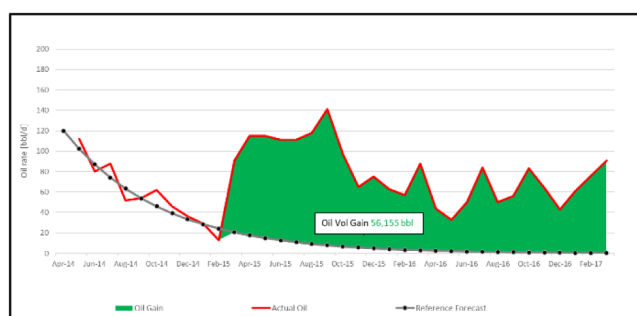


Figure 17—Post-stimulation production trends for Natih, Well 23.

In each case, actual production outperformed the no-stimulation forecast, confirming the LVA program's ability to extend productive life and increase ultimate recovery in the Natih Formation. While these early results are highly encouraging, some rate fluctuations suggest evolving flow redistribution within the multilayer Natih interval. Continued monitoring and potential targeted interventions—such as selective CT notching—could further optimize layer contribution and sustain the stimulation benefits.

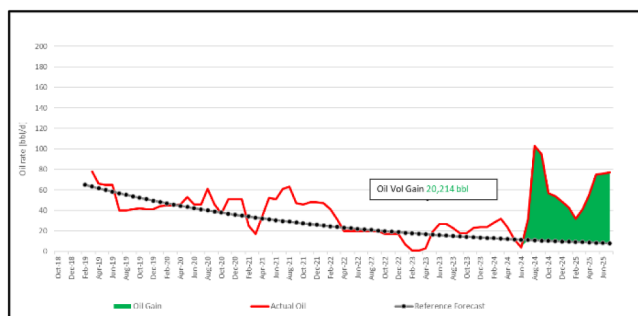


Figure 18—Post-stimulation production trends for Natih, Well 32.

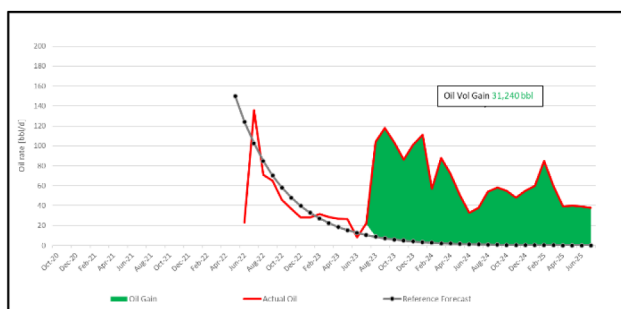


Figure 19—Post-stimulation production trends for Natih, Well 24.

Shuaiba Formation. The pre-job production history shows a high initial oil rate that declined steeply and then flat-lined at a low level. Post-stimulation, the well displayed an immediate and pronounced jump in liquid and oil output—roughly a four-fold increase relative to the pre-job plateau. Importantly, the elevated rate has been sustained over several months with minimal drawdown fluctuations, indicating that the retarded acid system achieved deep penetration through the relatively homogeneous Shuaiba matrix. While the early results are promising, ongoing pressure-rate analysis is recommended to assess the longevity of the uplift. As with Natih, selective CT notching could be deployed in future campaigns to ensure that any bypassed sections of the single Shuaiba layer are fully accessed. In Figure 20, Well 19's oil rate increased from ~80 BOPD to nearly 120 BOPD immediately post-stimulation, maintaining >100 BOPD for months and delivering a cumulative oil gain of 27,360 bbl.

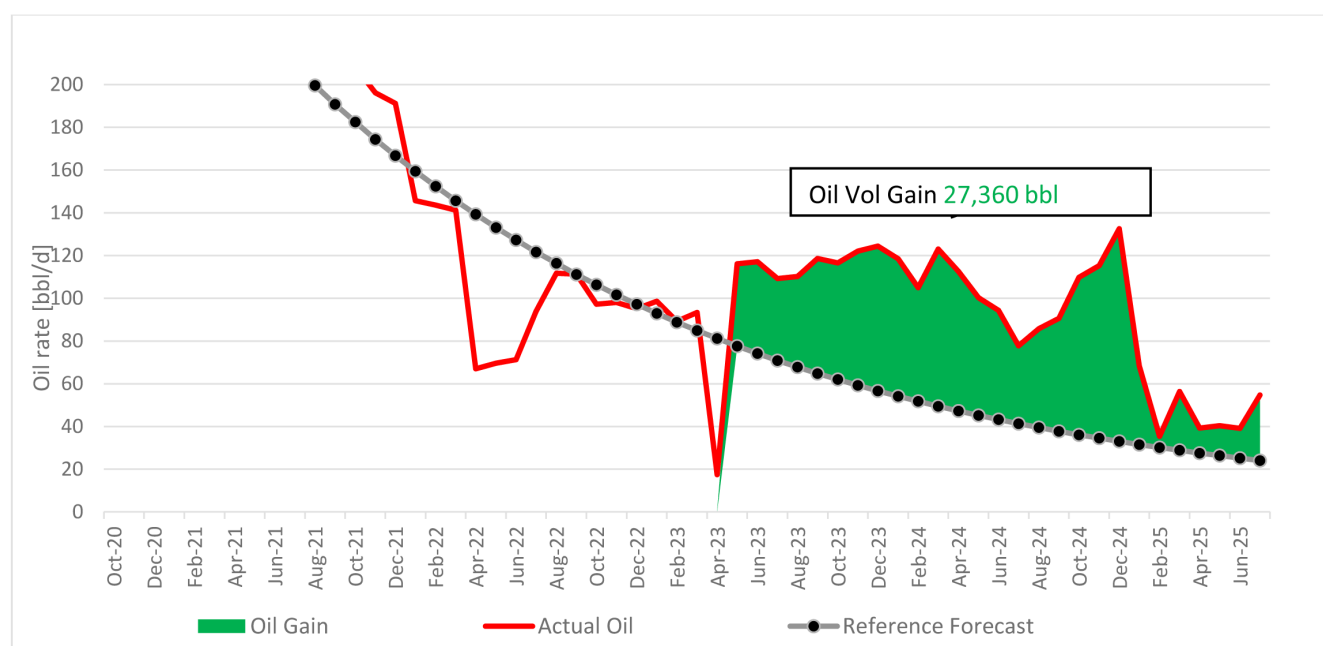


Figure 20—Post-stimulation production trends for Shuaiba, Well 19.

Future Challenges

Although the large-volume, rig-less acid program has delivered substantial value, several forward-looking challenges must be addressed to sustain and enhance its impact.

- **Diminishing "low-hanging-fruit" opportunities** – Most easily accessible Natih candidates have already been treated, leaving a higher proportion of thinner, tighter Shuaiba and Shammar intervals. This shift requires recalibrating the candidate-ranking matrix and staging workflow to reflect the higher risk-to-reward profile, with greater emphasis on diversion potential, localized damage indicators, and economic thresholds suitable for marginal reservoirs.
- **Expanding the stimulation toolkit** – Results from some Shuaiba and Shammar wells indicate that a purely rig-less LVA approach can be insufficient where pay is thin or laterally discontinuous. Future campaigns should evaluate coiled-tubing notching with targeted acid placement, or hybrid acid/prop-fracture designs, to improve interval-by-interval coverage and initiate deeper fractures in ultra-low-permeability zones.
- **Strengthening surveillance and zonal diagnostics** – Current monitoring, based mainly on bulk production trends, cannot definitively identify contributing zones or bypassed intervals. A more comprehensive surveillance program—incorporating production logging tools (PLT), chemical tracers, and rate–pressure diagnostics—will help quantify post-treatment inflow distribution and refine diversion strategies.
- **Real-time placement measurement** – The 2025 pilot of High-Frequency Pressure Monitoring (HFPM) has shown promise, but further integration with well production and surveillance data is needed. In future deployments, HFPM should be embedded within a closed-loop execution model, enabling diagnostic feedback to guide dynamic adjustments to pumping schedules, diversion sequencing, and rate transitions in real time.

By advancing candidate selection, stimulation methodology, surveillance capabilities, and real-time diagnostics, the LVA workflow can be adapted to increasingly challenging reservoirs. Over time, HFPM could become a standard component of LVA operations in Daleel, bridging current surveillance gaps

and enabling data-driven optimization to improve stage coverage, stimulation efficiency, and treatment consistency across Block 5's heterogeneous carbonate formations.

Summary

The 2025 Large-Volume Acidizing (LVA) campaign in Block 5 introduced two major advancements — the Next-Generation Single-Phase Retarded Acid (NG-SPRA) and High-Frequency Pressure Monitoring (HFPM). NG-SPRA delivered deeper, more uniform wormhole penetration and higher stimulation efficiency than the 2024 SPRA campaign, while HFPM provided the first real-time, depth-specific fluid placement diagnostics in Block 5, paving the way for a future closed-loop design–execution–feedback workflow.

NG-SPRA improved average gross FOI by ~4.6 and oil FOI by ~4.1 in Natih wells — 2.5–3× higher than the 2024 SPRA results. The approach maintained or improved stimulation coverage in 1,000–1,500 m open-hole horizontals without the use of coiled tubing, achieving up to 85% cost savings compared with multistage completion alternatives.

While these results strongly indicate NG-SPRA's potential to enhance productivity in heterogeneous carbonates, they are based on a small dataset and short observation period, with no normalization for reservoir quality (kh) or completion differences. Future work will focus on expanding HFPM integration for real-time pumping adjustments, incorporating PLT, tracers, and rate–pressure diagnostics to refine diversion strategies, and adapting designs for thinner, tighter Shuaiba and Shammam intervals using hybrid or targeted treatments.

This work establishes a scalable, data-driven carbonate stimulation methodology that can be transferred to other Omani fields and comparable reservoirs worldwide.

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